



NEPAL COLLEGE OF INFORMATION TECHNOLOGY

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LAB REPORT

Data Science and Analytics

Lab 03

Data Cleaning and Preliminary Visualization

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1. Objective

The objectives of this experiment are:

- To create and inspect a pandas DataFrame.
- To identify missing values and duplicate records.
- To perform data cleaning using imputation techniques.
- To remove duplicate records and outliers.
- To standardize numerical data.
- To perform data transformation.
- To visualize data using various charts.
- To analyze the relationship between experience and salary.

2. Theory

Data cleaning is the process of identifying and correcting inaccurate, incomplete, duplicate, or inconsistent data before analysis. Clean data ensures reliable insights and improves the quality of decision making.

Common data quality issues include:

- Missing Values
- Duplicate Records
- Invalid Values
- Inconsistent Data
- Outliers

Exploratory Data Analysis (EDA) uses statistical summaries and visualizations to understand patterns, trends, and relationships within data.

3. Software and Libraries Used

3.1 Software

- Python 3.x
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

3.2 Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
```

4. Dataset Description

The Employee dataset contains 15 employee records and includes:

- Employee ID
- Employee Name
- Department
- Age
- Experience
- Salary

The dataset intentionally contains:

- Missing values
- Duplicate rows
- Salary outlier (950000 NPR)
- Extreme age value (90 years)

5. Procedure and Results

5.1 Q1: Creating DataFrame

The dataset was loaded into a pandas DataFrame.

```
Shape of DataFrame:
(15, 6)

First Five Rows:
   EmpID  Name Department  Age  Experience  Salary
0   E01  Aarav         IT  28.0         4.0  45000.0
1   E02   Sita         HR  35.0         9.0  52000.0
2   E03  Bibek    Finance  NaN         6.0  48000.0
3   E04  Anita         IT  41.0        15.0  70000.0
4   E05  Rajan  Marketing  30.0         5.0     NaN

DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   EmpID           15 non-null    str
 1   Name            15 non-null    str
 2   Department      14 non-null    str
 3   Age             14 non-null    float64
 4   Experience       14 non-null    float64
 5   Salary          14 non-null    float64
dtypes: float64(3), str(3)
memory usage: 852.0 bytes
```

Figure 1: Output of df.head()

Observation:

- Dataset contains 15 records.
- Dataset contains 6 columns.

5.2 Q2: Missing Values and Duplicate Records

Missing values and duplicate records were identified.

```

Missing Values Per Column:
EmpID      0
Name       0
Department  1
Age        1
Experience  1
Salary     1
dtype: int64

Number of Duplicate Rows:
2

Duplicate Rows:
   EmpID  Name Department  Age  Experience  Salary
12   E04  Anita         IT  41.0         15.0  70000.0
14   E06   Puja         HR  26.0          2.0  38000.0

```

Figure 2: Missing Values Count

Observation:

- Missing values exist in Age, Experience, Salary and Department.
- Duplicate rows were detected.

5.3 Q3: Imputing Missing Values

Median imputation was used for Age, Experience and Salary.

Mode imputation was used for Department.

Observation:

No missing values remained after imputation.

5.4 Q4: Cleaning Wrong Data

The following operations were performed:

- Duplicate removal
- Outlier removal using IQR
- Removal of Age = 90

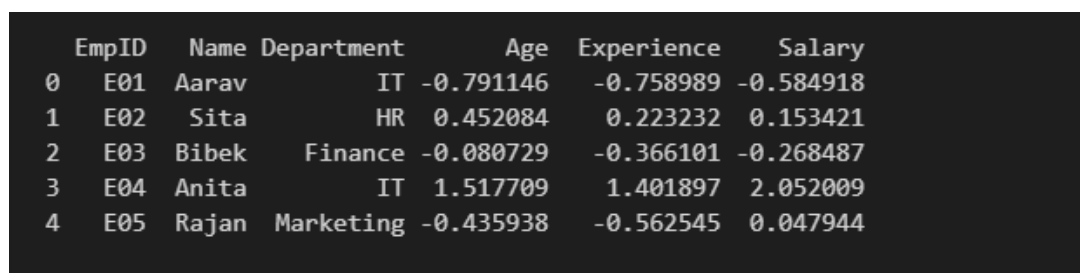
Observation:

Salary value 950000 NPR was identified as an extreme outlier.

5.5 Q5: Standardization

Z-score standardization was applied:

$$z = \frac{x - \mu}{\sigma}$$



	EmpID	Name	Department	Age	Experience	Salary
0	E01	Aarav	IT	-0.791146	-0.758989	-0.584918
1	E02	Sita	HR	0.452084	0.223232	0.153421
2	E03	Bibek	Finance	-0.080729	-0.366101	-0.268487
3	E04	Anita	IT	1.517709	1.401897	2.052009
4	E05	Rajan	Marketing	-0.435938	-0.562545	0.047944

Figure 3: Standardized Dataset

Observation:

All numerical columns were standardized successfully.

5.6 Q6: Data Transformation

The following transformations were performed:

1. Log Transformation of Salary
2. Label Encoding of Department
3. Creation of Age Groups

Observation:

The transformed dataset became more suitable for analysis.

5.7 Q7: Histogram of Age

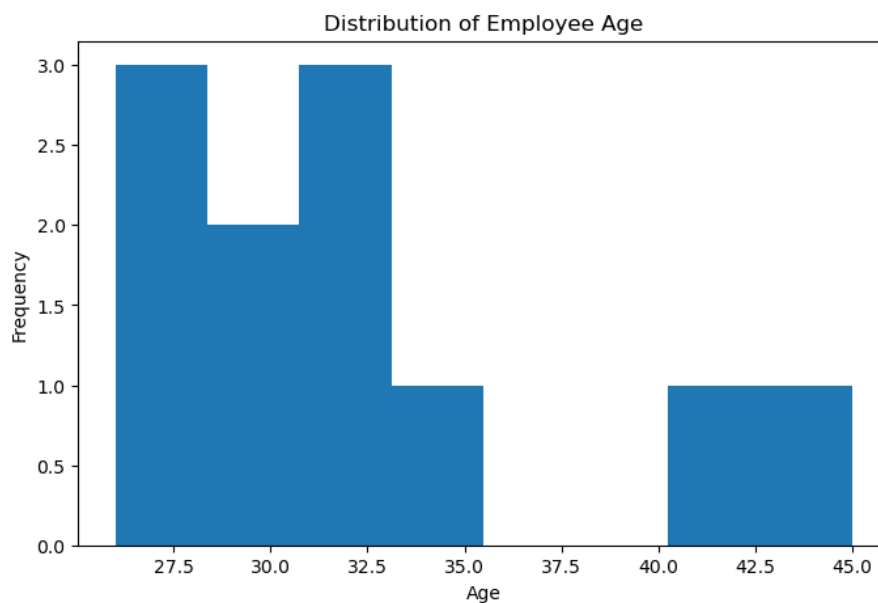


Figure 4: Histogram of Age Distribution

Observation:

Most employees belong to the age group between 25 and 40 years.

5.8 Q8: Box Plot of Salary

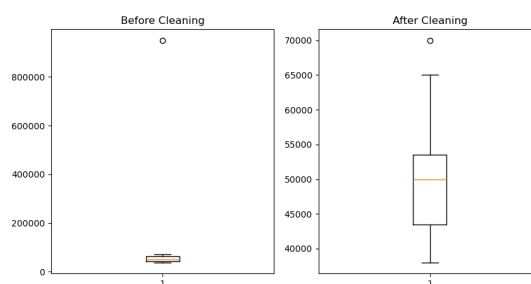


Figure 5: Salary Before and After Outlier Removal

Observation:

Salary value 950000 NPR appeared as an extreme outlier and was removed.

5.9 Q9: Department Count Plot

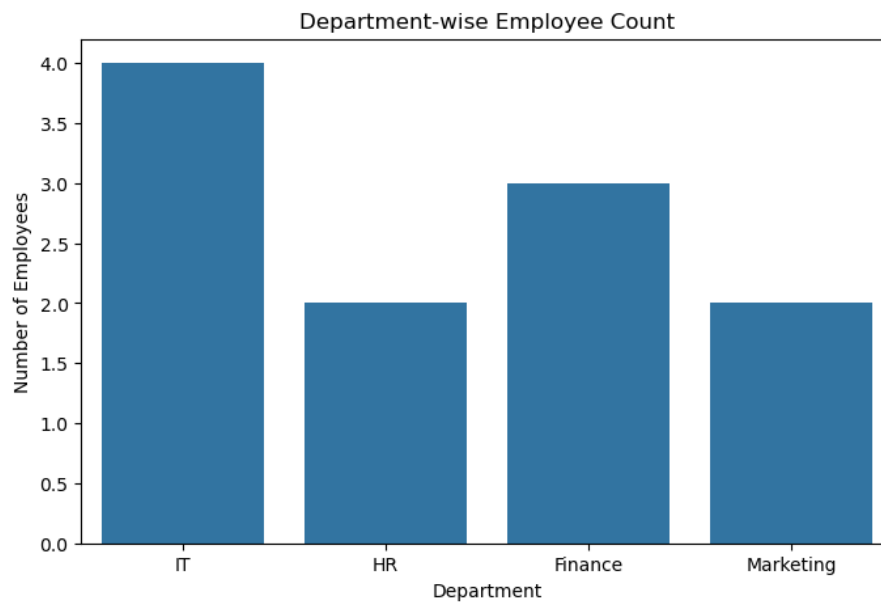


Figure 6: Department-wise Employee Count

Observation:

The IT department contains the highest number of employees.

5.10 Q10: Scatter Plot of Experience vs Salary



Figure 7: Experience vs Salary

Observation:

A positive relationship exists between experience and salary.
Employees with greater experience generally receive higher salaries.

6. Results

The following results were obtained:

- Missing values handled successfully.
- Duplicate records removed.
- Outliers detected and removed.
- Data standardized successfully.
- Visualizations generated successfully.

Key Findings:

- IT department has the highest number of employees.
- Most employees belong to the age group 25–40 years.
- Salary increases with experience.
- Salary outlier significantly distorted the original distribution.

7. Conclusion

This experiment demonstrated practical data cleaning and exploratory data analysis techniques using Python. Missing values were handled using statistical imputation, duplicate records were removed, and outliers were identified using the IQR method. Several visualizations were created to understand employee demographics and salary trends. The analysis showed a positive relationship between experience and salary and highlighted the importance of preprocessing before performing data analysis.

8. Viva Questions

1. What is data cleaning?
2. Why is median preferred over mean for imputation?
3. What is an outlier?
4. What is the IQR method?

5. What is z-score standardization?
6. What relationship was observed between experience and salary?